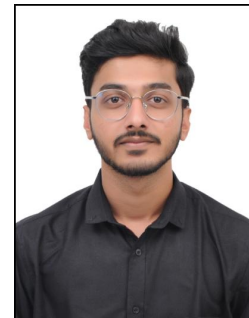


Lohith Tarikere Prasanna

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[LinkedIn](#) | [GitHub](#)

International M.Sc. student | eligible to work in Germany up to 140 days/year | open to Werkstudent & internship roles



Profile

M.Sc. Electromobility student (AI & Autonomous Driving) at FAU Erlangen-Nuremberg. Built and deployed a production AI tool at National Instruments that manufacturing engineers use daily to query measurement data and classify test results automatically. Strengths in Python data pipelines (Pandas/Polars), ML-based anomaly detection, and embedded automotive systems (STM32, CAN).

Experience

National Instruments GmbH (NI, now part of Emerson)

Dresden, DE

AI Workflows in RF Measurements — Intern

10.2025 – 03.2026

- Built and deployed an AI analysis tool now used daily by manufacturing engineers at NI Dresden to query production yield data and receive automated results, removing manual reporting from the workflow.
- Designed a Euclidean-distance ranking algorithm to surface failure patterns in production measurement data; scaled it to 30–40 measurement types without changing the existing data architecture.
- Built an LLM agent skill that routes natural-language queries to the correct Python analysis pipeline and returns structured results with embedded Plotly charts, letting engineers query production data without writing code.
- Benchmarked Polars against pandas on 100k+ row datasets; measured a 5–10× latency gain that drove full adoption, and added Parquet caching to eliminate redundant datastore reads.
- Implemented a custom $1\sigma/3\sigma$ statistical classifier to split test outcomes into genuine failures, marginal cases, and noise, with no external ML dependencies.
- Owned end-to-end integration into NI's proprietary datastore: schema design, API contracts, and backward compatibility across legacy data formats.

Delta X Automotive Pvt. Ltd.

Bengaluru, IN

Embedded Systems Intern

12.2022 – 08.2023

- Programmed STM32 microcontrollers in embedded C for EV powertrain control, cutting system startup time by 25%.
- Integrated FreeRTOS on CAN-based ECUs, improving task scheduling and system reliability.
- Validated LV/HV architecture integration with hardware engineers using Simulink.

Projects

RF Modulation Classifier [\[GitHub ↗\]](#)

Jun 2026

- Built a classifier that recovers how a radio signal was encoded from raw waveform data alone, outputting the modulation type (e.g. BPSK, QAM) from 16 statistical features spanning amplitude, phase, frequency, and higher-order spectral patterns.
- Trained and evaluated on RML2016.10a (220k-sample benchmark, 11 signal types, 20 noise levels); reached 83.6% accuracy on clear signals vs. a 9% random baseline, with custom features lifting overall accuracy from 47.7% to 51.3%.

Intelligent Decision-Making for Cognitive ISAC Networks

Project Thesis, ongoing

- Identified a gap in ISAC literature: radar cross-section is modelled as an i.i.d. random variable, when in distributed multi-node networks it is geometry-dependent; proposing a deterministic bistatic RCS model $\sigma(\theta_b)$ that makes node-role selection a meaningful optimization variable.
- Formulating a joint node-role selection and beamforming problem for a multi-node V2X intersection ISAC network, solved via convex optimization and successive convex approximation (SCA).

Visible Light Communication System for EV V2V Applications

- Built an STM32-based Vehicle-to-Vehicle communication prototype using OOK optical modulation with Turbo Coding for forward error correction.
- Simulated the LED driver in Multisim for voltage-stability validation and modelled an MPPT algorithm in MATLAB for a solar-powered transceiver.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

Erlangen-Nürnberg, DE

M.Sc. Electromobility — Majors: AI & Autonomous Driving, Connectivity

04.2024 – present

Focus areas: MIMO Systems, 5G/6G Architectures, DSP Architectures, Energy Informatics & Smart Grids

Dayananda Sagar College of Engineering

Bengaluru, IN

B.E. Electronics and Communication Engineering

08.2019 – 08.2023

Skills

Programming & Data: Python (Pandas, Polars, scikit-learn, Plotly), Embedded C, MATLAB

Machine Learning: anomaly detection, statistical classification, feature engineering, large-scale data processing, Anthropic SDK / agent tooling

Embedded & Automotive: STM32, FreeRTOS, CAN Bus, UART/SPI

Tools: Git, Azure DevOps, Parquet

Certification: Anthropic Academy — Model Context Protocol (MCP), 05.2026

Languages

English: C1 — Proficient (CEFR)

German: A2 — actively improving